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Nano-scale mechanical properties of constituent minerals in shales investigated by combined nanoindentation statistical analyses and SEM-EDS-XRD techniques

Yiwei Liu [a,b,d], Ang Liu [b], Shimin Liu [b], Yong Kang [c,d,*]

[a] School of Civil Engineering, Wuhan University, Wuhan, 430072, Hubei, China
[b] Department of Energy and Mineral Engineering, G[3] Center and EMS Energy Institute, The Pennsylvania State University, University Park, PA, 16802, USA
[c] School of Power and Mechanical Engineering, Wuhan University, Wuhan, 430072, Hubei, China
[d] Hubei Key Laboratory of Waterjet Theory and New Technology, Wuhan University, Wuhan, 430072, China

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ABSTRACT

The mechanical properties of the constituent minerals in shale rock are fundamental to a better knowledge of multi-scale shale behaviors. It benefits the engineering applications and predictive physics modeling of the shale formation. In this work, nanoindentation testing combined with scanning electron microscopy - energy dispersive spectroscopy (SEM-EDS) were used to obtain the *in situ* mechanical properties of individual mineral phases in four shale samples. Engraved cross-marks were employed to locate the indented areas on a micron scale for subsequent SEM-EDS measurements and analysis. The elastic moduli and hardness of the quartz, iron-type minerals, muscovite, clinochlore, organic matter, and the mineral assortments in the matrix were analyzed and relevant deformative behaviors were compared. Results show that the identified mineral groups exhibit a diverse set of mechanical properties at the nanoscale. The iron-type minerals have the highest elastic modulus (104.7 GPa), then followed by quartz, muscovite, clinochlore, and organic matter. Quartz shows the highest hardness (10.2 GPa). The identified minerals demonstrated different elasto-plastic characteristics to the indentation loads. Quartz exhibits the lowest plastic behavior, while phyllosilicates exhibited large plastic behavior owing to the layered structure. Organic matter showed both elastic-dominant and plastic-dominant behaviors, which may be related to the chemical compositional differences and various thermal maturity of kerogen in the shale samples. The mechanical properties of three mineral assortments in shale matrices vary significantly due to the wide variety of the mineral compositions. Radial cracks were observed on the boundary of brittle minerals and relatively weak minerals, while shear cracks were observed on strength-weak minerals such as layered silicates and organic matters. The mechanism of the induced cracks was discussed. This study obtains a basic understanding of shale behavior on nano-scale and provides reliable basic data for multi-scale modeling of shale reservoirs without the need for large-scale mechanical tests.

1. Introduction

The mechanical behaviors of the shale play a crucial role in geoscience topics and engineering applications, such as hydrocarbon extraction, carbon sequestration and storage, and ground control in underground coal mines.[1–6] The elastic property and strength of shale rock strongly influence the geometry of fractures induced in the reservoir stimulation, and the transport behavior of gas or oil liquids, further impacting the economic benefits of the hydrocarbon extraction.[7,8] Shale is the common sealing cap-rock in hydrocarbon reservoir and roof in coal mines, its mechanical behavior also firmly affect the potential for carbon dioxide geological storage in depleted hydrocarbon reservoirs and drained coal mines.[2,9,10] Besides, the roof failure of the underground coal mines is a problematic issue in the ground control safety, which involves the weak behavior of the shale roof.[6,11] All these energy exploitation applications and decarbonized attempts give rise to the need for a deeper understanding of shale behaviors. Based on the lithologic observations of shale microstructure, it is reported that shale is formed by the compaction of mechanically various particles like quartz, clay minerals, and organic matter.[12,13] Mineralogical composition is an

* Corresponding author. Hubei Key Laboratory of Waterjet Theory and New Technology, Wuhan University, Wuhan, 430072, China.
E-mail addresses: szl3@psu.edu (S. Liu), kangyong@whu.edu.cn (Y. Kang).

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important factor that influences the mechanical property of shale.[14] The understanding of the mechanical properties of the constituent minerals of shale is fundamental to a better knowledge of multi-scale shale behaviors.

Mechanical testing has been widely conducted on shales to gain insight into the mechanical properties of shales in past decades, such as uniaxial or triaxial compressive strength tests, shear tests, and ultrasonic measurements.[15–19] These commonly used meso- and/or macroscopic laboratory tests provide the bulk properties of shale mass, but are impractical to measure the mechanical properties of the individual mineral components due to the small size of the mineral grains.[20] The nano-scale mechanical behaviors of the mineral components are important to the cross-scale data integration and predictive physics modeling of shale for both fluid transport and failure behavior.[21–23]The spatial distributions of these components are of great significance for the prediction and evaluation of the resource potential of shale formations. The properties of the fine-particle porous clays and diagenetic minerals are intrinsically controlled by single mineral and their distributions on the nanometer to micrometer scale. To date, the *in situ* mechanical responses of single mineral phase in shale remain incompletely understood.

The improvement in sensor technology and high-resolution imaging allows the acquisition of material properties on small scales. Nanoindentation is a commonly used technique for mechanical testing in material the science field in the last decades. Owing to the ability to reveal the properties of individual components under prescribed loads, it has been performed to characterize the properties of different materials at the submicron scale before.[24–29] A typical nanoindentation process usually involved an indenter with a diamond tip, intruding the tested material under specific load or displacement control, and then withdrawing from the material, leaving a residual impression on the surface. It can obtain important mechanical parameters such as reduced modulus and hardness without large demand for rock samples and time-consuming testing procedure. From an operational perspective, nanoindentation testing is non-destructive to the sample, and it is fast in the testing period compared with conventional testing methods.[20,30] The precise indenter tip makes it possible to the locally mechanical characterization of constituent minerals in shale.[31] Another technique that is commonly used for nano-scale mechanical characterization on rocks is atomic force microscopy (AFM). It can provide three-dimensional topographical images of the target surface and mechanical mapping of the tested sample.[32–34]

Generally, repetitive nanoindentation tests are required to provide sufficient information about the mineral phases given the inherent heterogeneity of shale. In previous studies, grid nanoindentation tests and statistic deconvolution methods were adopted to extract the mechanical properties of shale.[12,35–42] Statistical analysis methods such as Probability Density Function (PDF), Gaussian Distribution Model, and Cluster algorithm, were used to analysze the indentation data and identify the mineral groups with different strengths and stiffnesses.[28,35,37,43–48] However, the classification of minerals groups was purely based on the multi-model behavior of the probing data. Direct evidence is scarce for linking indentation properties to their corresponding mineral phases. Another way to investigate the mechanical properties of heterogeneous material is to combine the nanoindentation tests with high-resolution imaging, such as optical microscopic, scanning electron microscopy (SEM), back-scattered electron diffraction (BSED), and energy-dispersive X-ray spectroscopy (EDS), which help locate the indents in the area of interest, detect the element distribution and identify the mineralogy at indentation sites.[28,39,45–51] Abedi et al. introduced a method that characterized the chemo-mechanical behavior of shale rock on a submicron length scale by combining nanoindentation data with EDS analysis.[51] Veytskin et al. combined nanoindentation with SEM-EDS to obtain the elastic modulus and hardness distribution of statistically distinguishable mineral phases.[37] Kumar studied the mechanical behavior of kerogen by combining nanoindentation and SEM-EDS and found that the increase of TOC decreased Young's modulus of shale.[39] Mason investigated the influence of lithofacies on the mechanical properties of shale using Vickers indentation combined with SEM-EDS analysis.[52] Yang et al. identified the minerals in shale matrix under an optical microscope and conducted nanoindentation tests on the specific minerals with different indentation loads.[47] Some limited studies have been devoted to investigating the properties of the mineral constituents in shale,[37,52,53] but the systematic data is still scarce and the nano-scale properties of the mineral compositions beyond skeletal particles (such as the mixture part in the matrix) are ignored. Hence, further studies are essential for a definitive understanding of the mineral-dependent property variations.

In this study, we conducted nanoindentation tests on four shale samples with various geological characteristics to determine the *in situ* mechanical properties (elastic moduli and hardness). The residual impressions on the shale surface were directly observed by scanning electron microscopy (SEM), and the element compositions were determined by energy-dispersive X-ray spectroscopy (EDS). X-ray diffraction (XRD) was used to analyze the bulk mineral compositions. The representative mineral groups were identified combine the probing data with SEM-EDS-XRD results, and then obtained their mechanical properties and elasto-plastic behaviors. The mineral assortments in the matrices, which are distinct between different shales, are mechanically characterized. The outcome of this study can directly link the lithologic and microstructural information to the mechanical behaviors of shale, as illustrated in Fig. 1. Potentially, the method and results can provide reliable input data for multi-scale, geomechanically modeling of shale reservoirs.

2. Experiment procedure

2.1. Samples and sample preparation

Three black shale samples and one gray shale sample were collected: Illinois Energy Shale 1 (IL-1), Appalachia Shale 2 (AS-2), Appalachia Shale 3 (AS-3), and Appalachia Shale 1 (AS-1). The sampling locations are denoted as yellow stars in Fig. 2. Sample IL-1 was recovered from Herrin coalbed at Upper Carbondale Formation in Illinois, and the other three samples were collected near Appalachian Mountain Section, Pennsylvania. Thereinto, sample AS-1 was collected from the Lower Kittanning Formation in Clearfield County, while samples AS-2 and AS-3 were collected from Clarian Formation and Upper Freeport Formation at Allegheny Group in Indiana County.

The shale samples were first cut into small wafers which can be

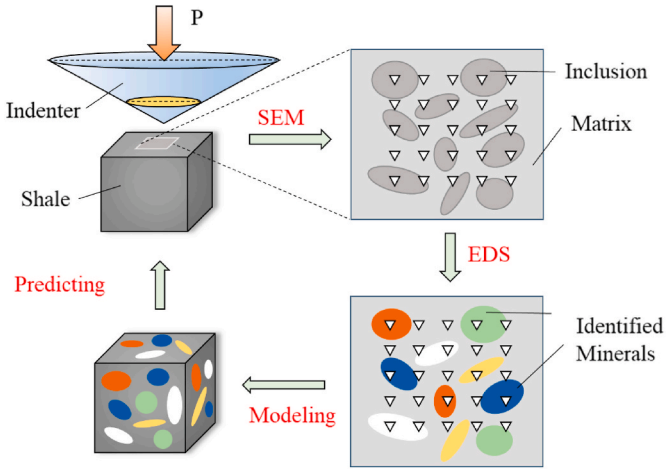


Fig. 1. Schematic of the identification of constituent minerals in shale by combining nanoindentation with SEM-EDS techniques, which paths a way to a better evaluation of shale's mechanic behaviors on a bulk scale.

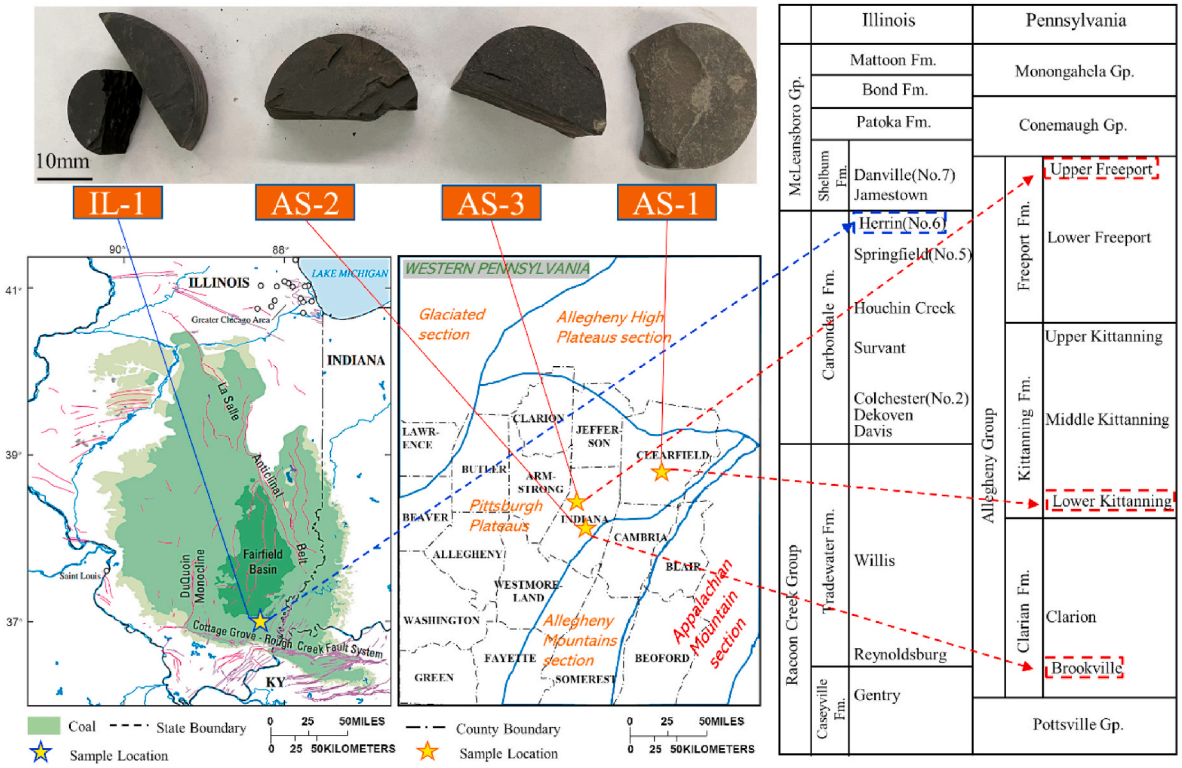


Fig. 2. Sampling locations of the four shale samples.[54–56]

properly mounted for nanoindentation test, as shown in Fig. 3(a). After wafer preparation, the residual shale fragments were grinded to powder less than 325 mesh and used for X-ray diffraction (XRD) analysis. Initially, all prepared samples were oven-dried for 24 h at 80 °*C*. The main challenge of nanoindentation on shale, known as a highly heterogeneous material, is to develop an appropriate sample preparation procedure to minimize the impact of surface roughness. The indenter tip, as shown in Fig. 3(c), is very precise and sensitive to the surface roughness of the tested sample. Thus, polishing in precision is required and necessary. The shale samples were first cast in epoxy resin prior to the polishing to keep the integrity during the preparation. All surfaces ready for polishing were fixed parallel to the bedding planes. The epoxy was lower than the surface of the samples to avoid contamination. Then, grinding and polishing were performed with abrasive papers of different grain fineness levels and diamond suspensions.[57] Broad ion beam milling was also adopted, which is a sample preparation technique in geomaterials to achieve a great surface flatness.[58–60] Fig. 3(b) shows the flat and smooth surface of the shale samples after polishing. The root mean squared (RMS) roughness of the shale samples is about 120–150 nm which is measured by an optical profilometer, as shown in Fig. 3(d).

Surface roughness is a key factor influencing the accuracy of the measured data in nanoindentation. Since the contact area of the indenter and the tested sample is indirectly measured, high surface roughness can cause inaccurate estimation of the contact area and leads to overestimation or underestimation of probing mechanic properties.[61] It was recommended that the maximum indentation depth be 3 times larger than the RMS roughness of porous materials.[62] Different from the synthetic materials, it is hard to mill the natural rock down to 100 nm due to the discrepancy in properties between their mineral constituents.[20,63,64] The impact of surface roughness can be reduced by conducting the tests

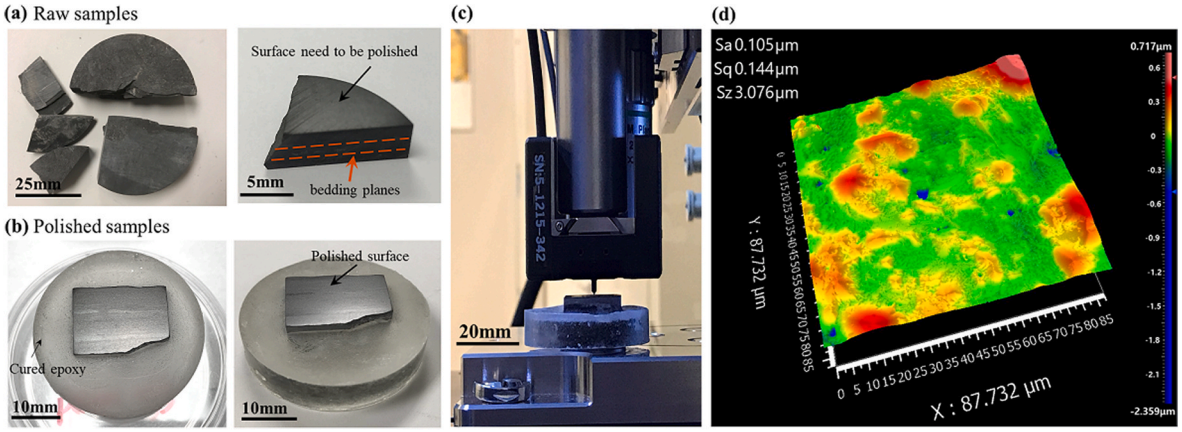


Fig. 3. Sample preparation before the actual nanoindentation test. (a) The raw samples were cut into small wafer pieces. (b) The sample surface was flat and shining after polishing using abrasive paper and diamond suspension. It was cast into epoxy to keep it from spalling or damaging during the polishing. (c) Indenter tip on a prepared sample. (d) Surface roughness measurement conducted by an optical profilometer.

on smoother areas or applying higher loads. We examined the indentation locus one by one on the SEM images and those that fell on pores or locally rough surfaces were excluded from the data analysis. This can significantly diminish the impact of surface roughness and ensures the accuracy of the tested data.

2.2. Mineral composition analysis for bulk samples using XRD

Quantitative mineralogical information was achieved by XRD measurement for all samples. The mineralogical compositions were measured by RIGAKUD/Max-3B diffractometer. Small shale chips were smashed down to 325 mesh powder for the measurement. The measurement was conducted under vacuum and room temperature and the scan angle ranges from 5° to 70°. To allow quantification of the amorphous phase in the shale samples, 15%wt ZnO was added to the samples before each measurement. Jade software was used for the interpretation of the raw diffraction data and the determination of the mineral contents. The obtained XRD mineralogical information were applied as the reference to identify the minerals in the field of observation under SEM by using the elemental information under EDS.

2.3. Nanoindentation testing

The nanoindentation measurements were conducted on the Hysitron TI 980 TriboIndenter (Bruker Corp., US). An advanced accelerated property mapping mode (XPM accelerate mapping mode) was adopted in the nanoindentation testing. Through the coupling of a high bandwidth electrostatically actuated transducer, fast control electronics, and data acquisition, this new nanoindentation technique can perform 6 measurements per second. It can achieve comprehensive quantitative nanomechanical properties with high measurement resolution and accuracy in a short time, and much faster than traditional nanoindentation methodologies. During each run, an indenter with a precise diamond tip intrudes the material under the given load and then withdraws from the material, leaving a residual impression on the surface, as shown in Fig. 4 (a). A three-sided pyramid-shaped Berkovich intender tip was used to conduct the test. The load function adopted was shown in Fig. 4(b), which consists of 0.5 s linear loading to a maximum load of 10 mN, 0.2 s holding, and finally 0.5 s unloading to zero load. Considering that the indented areas are micron-scale and hard to find without any reference location, the cross marks were engraved on the samples by a thin blade and each testing sample was divided into four regions (upper right corner, upper left corner, lower left corner, and lower right corner), as shown in Fig. 4(c). The cross marks are about 100 μm wide. The distance between indentation grids to the cross-marked area is 2–3 times the width of the cross marks. Previous numerical studies on scratch tests show that the stress concentration due to the scratch is limited and only impacts the distribution of stress within the area which has a distance equivalent to the width of the scratch.[65,66] Thus, the additional stress resulting from the engraving does not impact the experiment results. With the engraved cross-marks, it is convenient to locate the indents for the subsequent SEM-EDS measurements and analyses. Nanoindentation tests were carried out in the four regions in the vicinity of the engraved mark on each sample. 8 by 8 array indents were conducted on each site, and the spacing was 10 μm. Totally 1024 nanoindentation tests were conducted on the four samples. All the load-displacement curves were recorded.

The elastic modulus and hardness of the tested shale can be obtained from the load-displacement curves using the method proposed by Oliver and Pharr[24] which are given by:

$$E = \left(1-\nu^2\right)\left[\frac{1}{E_r} - \frac{1-\nu_i^2}{E_i}\right]^{-1} \tag{1}$$

$$H = \frac{F_{max}}{A_c} \tag{2}$$

where E and ν are the Young's modulus and Poisson's ratio of the test specimen, while E_i and ν_i are Young's modulus and Poisson's ratio of the indenter tip. E_r is the reduced modulus that considers the effect of the non-rigid indenter and is given by[24,61]:

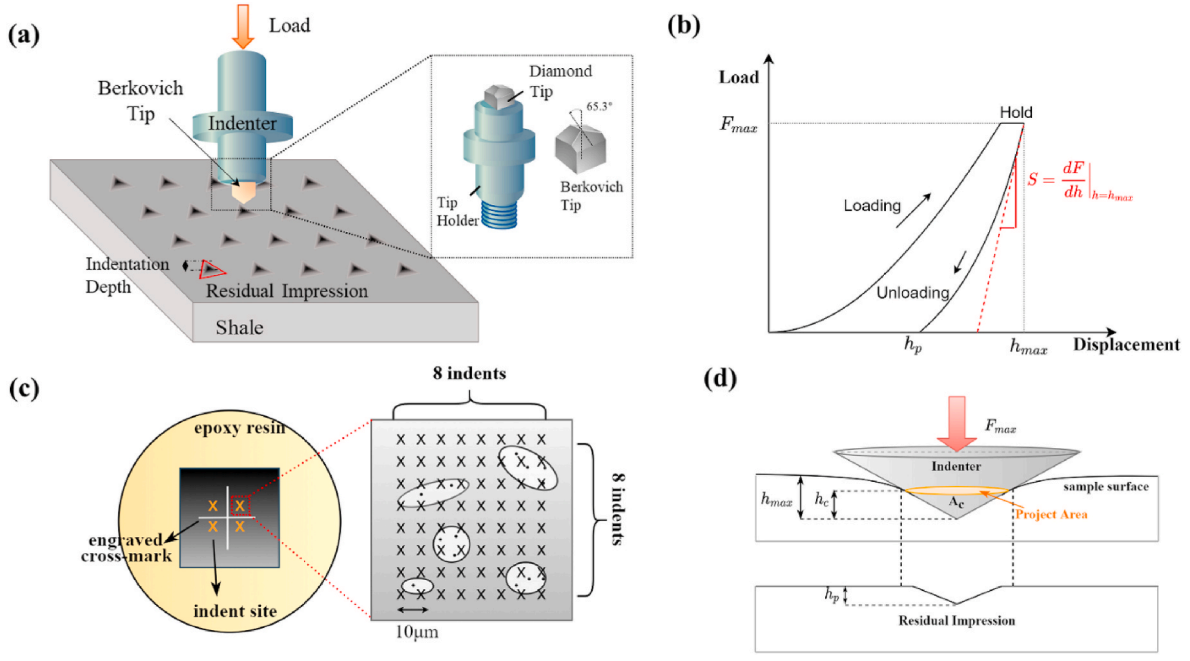


Fig. 4. (a) Schematic of the nanoindentation test, the indenter, and the Berkovich tip. The indenter with a Berkovich tip penetrated the shale samples and left a residual impression on the surface. The load-depth curves were recorded during this process. (b) Typical load-displacement curve in a nanoindentation test with loading, holding, and unloading process. (c) The samples were divided into four regions by the artificial engraved cross-marks. In each region, 64 tests were conducted with 10 μm spacing between two adjacent indents. (d) Schematic of the penetration by indenter tip and the residual impression.

$$S = \left.\frac{dF}{dh}\right|_{h=h_{max}} = \frac{2\beta\sqrt{A_c}}{\sqrt{\pi}}E_r \tag{3}$$

where S is the measured stiffness of the upper slope of the unloading curve, as shown in Fig. 4(b); β is a geometry correction factor and a value of 1.034 is commonly used for Berkovich indenter tip; h is the depth of indenter tip and h_{max} is the maximum depth. A_c is the project area which is a function of the contact depth h_c, as shown in Fig. 4(d).

In this study, a diamond tip was used in the nanoindentation tests, and the value is 1140 GPa for E_i, and 0.07 for ν_i. The Poisson's ratio varies for different mineral phases in practical. Fig. 5 shows the elastic modulus with different Poisson's ratios calculated by Equation (1). It is found that the differences of the elastic moduli are within 5% with a ± 0.1 value change in Poisson's ratios. Its impact on the estimation of Young's modulus is slight and the differences in the statistical elastic moduli of different phases are far more than the possible deviation caused by Poisson's ratio. This result agrees with previous studies which also suggest that Young's modulus is not sensitive when Poisson's ratio changes in a small range.[67,68] To offer comparable results with previous studies, a commonly used value of $\nu = 0.25$ was used in this work.

2.4. Scanning electron microscopy and elemental mapping

The scanning electron microscope (SEM) and energy dispersive X-ray spectroscopy (EDS) detection provide a quantitative and spatial mapping of the local mineralogy of the indented regions. The 2D morphology of the shale surface is qualitatively captured using SEM imaging, and the distributions of elemental compositions were quantified *via* EDS analysis. To obtain a high-resolution image of the shale sample, each sample was polished and then coated with iridium by avoiding heavy charging effects before SEM imaging. The acceleration voltage for the indented area was 10 kV. To obtain an appropriate EDS resolution and pertinent size, the acceleration voltage is normally adjusted to obtain the dead time between 10% and 30%. The system vacuum pressure is 2.75e-06 Torr and the gun vacuum pressure is 7.50e-10 Torr. The standard aperture size of 30 μm is used in this study. With the assistance of the engraved cross-mark, all the indents on the surface can be conveniently located after indentation and the indent morphology was captured by SEM mapping. Then we identified the integral and localized chemical elements present and quantified their abundance. This allows the determination of the representative mineral under the indents and correlated the indentation modulus and hardness with each mineral phase, as schematically shown in Fig. 6.

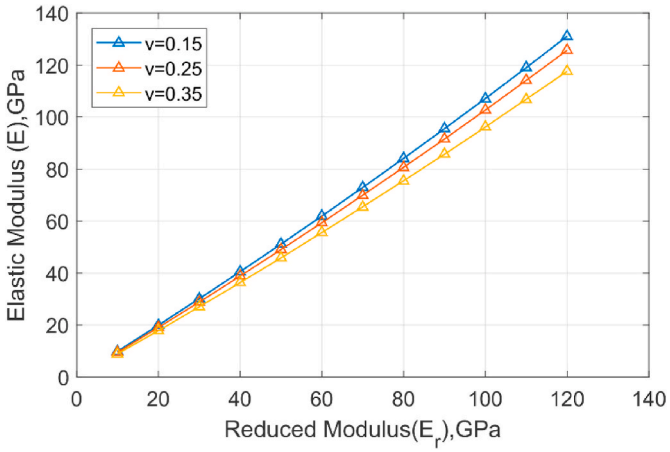


Fig. 5. The impact of different Poisson's ratios on the estimation of elastic modulus.

3. Results and discussions

3.1. Mineral composition of the samples

Table 1 lists the primary mineral compositions of the four shale samples used in our experimental program. It was found that quartz and layered silicate minerals were rich in the samples, and pyrite, siderite, and rutile are minor. The identified minerals in the samples can be divided into four groups: (1) Hard granular minerals, such as quartz, rutile, albite, and diopside; (2) Phyllosilicate minerals, including muscovite, clinochlore, kaolinite, and illite; (3) Iron-type minerals, which has smaller particle size than quartz. The iron-type minerals can be dispersed in the mineral assortment in the matrix (like siderite) or exist as framboids (like pyrite). (4) Organic matter (amorphous phase). Organic matter was found in black shale samples IL-1, and AS-2 through XRD analysis. In sample AS-3, organic matter was observed in the SEM and EDS images but was not detected in XRD. This may be attributed to both the low content of organic matter in AS-3 and the limited sample powder used for XRD analysis. Another shale sample from the same formation shows that an amorphous phase of 7 wt% is tested in XRD analysis. Thus, we took the results from SEM and EDS images of shale AS-3 and confirmed the presence of organic matter in it. No amorphous phase was detected in gray shale AS-1. The vitrinite reflectance (%R_0) value of the Marcellus Shale (AS-2, AS-3) in Indiana county is 1.3 to over 2.0, indicating a high thermal maturity, and the total organic carbon (TOC) content ranging from 1.5 to 6.2 wt%.[69] The black shale from Illinois Energy has a %R_0 as 0.71 which indicates a fairly low maturation, and the TOC is 21.4 wt% which is relatively high.[1] In addition to the above minerals, some associated minerals and rare metal minerals were observed in SEM-EDS analysis but failed to be identified by XRD. It is because their trace amounts are below the XRD detecting thresholds, such as ilmenite, chromite, apatite, and goethite. The results provide the basis for the identification of the indented mineral phases in SEM-EDS analysis. Noting that the mineral compositions in the areas selected for nanoindentation tests were unknown before the tests, and the mineral phases where the indents fell were random. This means that not all the minerals were captured by nanoindentation, especially the minerals with small fractions.

3.2. Surface imaging and elemental distribution analysis

With the assistance of the engraved cross-marks, it is convenient to locate all the indentation grids under SEM measurements. Fig. 7 shows the surface morphology of tested shales under the optical microscope after nanoindentation tests. It is recognized that shale samples contain a variety of coarse-grained and fine-grained particles with various alignment angles. The relatively large granules appeared as clusters in the field of observation with distinguishable geometry, shape, size, and spatial distribution. The spaces between the large granules were filled with fine-grained laminar minerals (such as in sample AS-1) or dark organic matters (such as in sample AS-3), as shown in Fig. 7. It is apparent that shale is highly heterogeneous and multiphase based on the direct observation. From Fig. 7, the permanent residual impressions can be observed after the probe is withdrawn from the specimen's surface. Since the nanoindentation tests were carried out under the same maximum load, the ultimate depths of the indents depend on the hardness of the material or material cluster being indented. Due to the heterogeneity of shale, the indents were measured to have various depths in the SEM images. It was found that a few indents fell on rough sites or pores although we cautiously selected more smooth areas for the nanoindentation tests. This is almost inevitable owing to the different removal rates and preferred polishing of the minerals in the shale. Hard minerals (such as quartz and feldspar) are easily pulled out of the weakly cemented clay matrix, leaving pores on the surface as shown in Fig. 7. In fact, some pores are naturally existed in shale and cannot be eliminated by the polishing procedure. To ensure the accuracy of the database, the

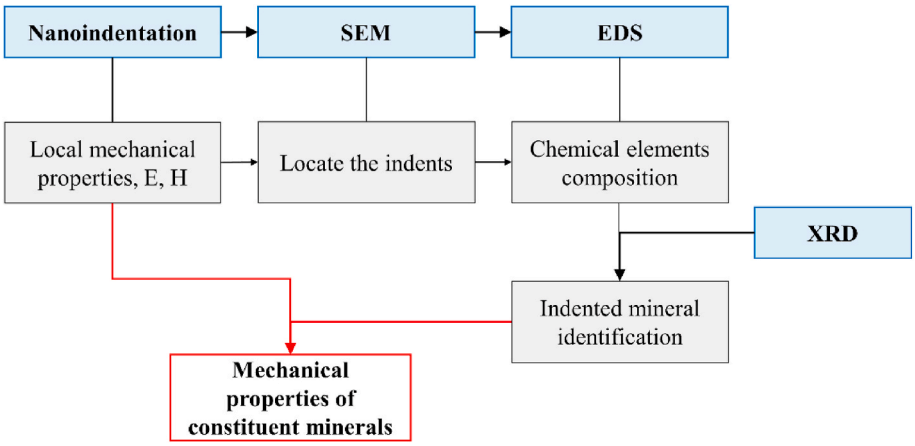


Fig. 6. Flowchart of analytical methods used in this study.

Table 1
Main minerals in the shale samples.

Minerals (wt%)	AS-1	AS-2	AS-3	IL-1
Quartz	32.6	23.4	26.9	11.7
Pyrite	0.3	0.3	–	1.6
Muscovite	45.5	–	30.5	17.0
Kaolinite	11.3	10.9	19.5	1.6
Clinochlore	9.5	9.1	16.7	–
Chlorite-serpentine	–	–	–	6.6
Illite	–	18.0	–	17.4
Amorphous	–	15.0	–	37.4
Siderite	–	8.2	3.2	–
Dolomite	–	–	–	3.1
Rutile	0.7	7.8	3.2	–
Diopside	–	6.0	–	
Albite	–	–	–	2.5

indents that fell on rough surfaces were discarded because the data can be significantly influenced by the localized morphological disparity.

Fig. 8 shows the elemental distributions of O, Al, Si, Mg, Fe, S, Na, K, and Ti based on the EDS-layered images. The major elements are shown in different colors, where luminance and contrast indicate the relative abundance of any specific element. O, Si and Al are the most abundant elements in all shale samples. Two highly-contrasted phases with a significant difference in grain size were observed in the EDS images: coarse grains and a fine-grained assortment in the matrix. The surface of sample AS-1 is mainly composed of large-area blue-green fine-grained minerals (rich in Mg, Na, Al, and K) and large size granules (rich in Si), which is consistent with the results of XRD that the sum of quartz and layered silicate minerals is up to 99%. Carbon-rich organic matters are observed in sample AS-2, AS-3, and IL-1. Samples IL-1 presents a large area of carbon-rich regions, corresponding to the high contents of amorphous substances in the XRD results. Large-grained Si-rich minerals appear as magenta blocky aggregates, which are almost isolated from each other or fully enfolded by the fine-grained matrix, indicating that no direct contact between these large grains during diagenesis. By matching the positions of SEM and EDS images, we marked the residual impressions on the EDS images illustrated by the white circles. The indents were found to be small enough to fall on a single phase like organic

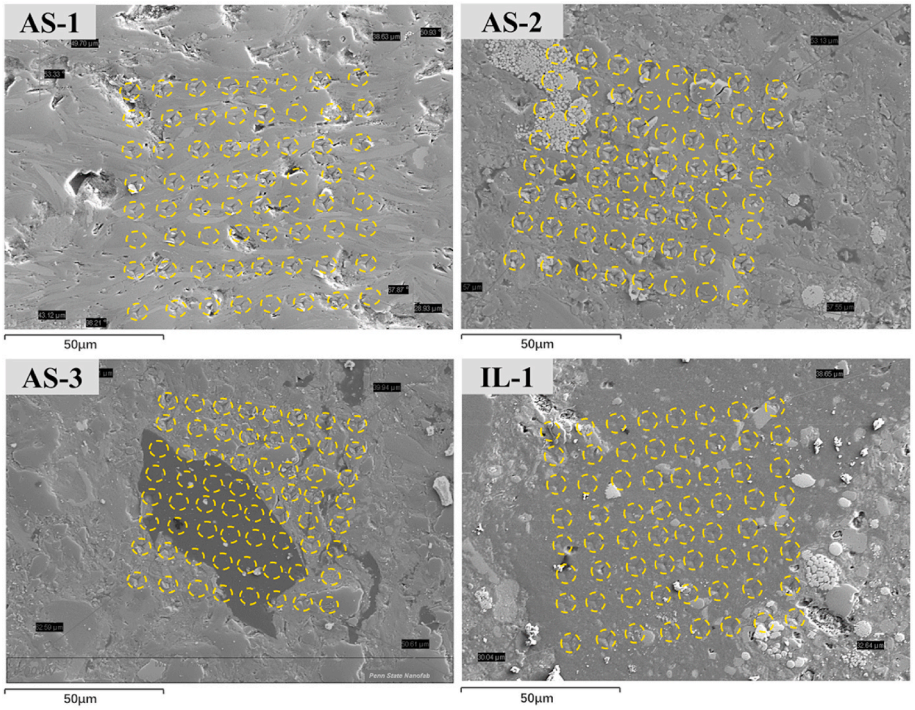


Fig. 7. Electron images from SEM for shale samples AS-1, AS-2, AS-3, IL-1. The yellow dot circles denote the residual impressions left by the indenter. Stripe or band-shaped minerals were observed in the sample AS-1. Granular grains were rich in sample AS-2. Organic matter was observed in sample AS-2, AS-3, and IL-1. The residual impressions vary in depth due to the heterogeneity of mechanical properties. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

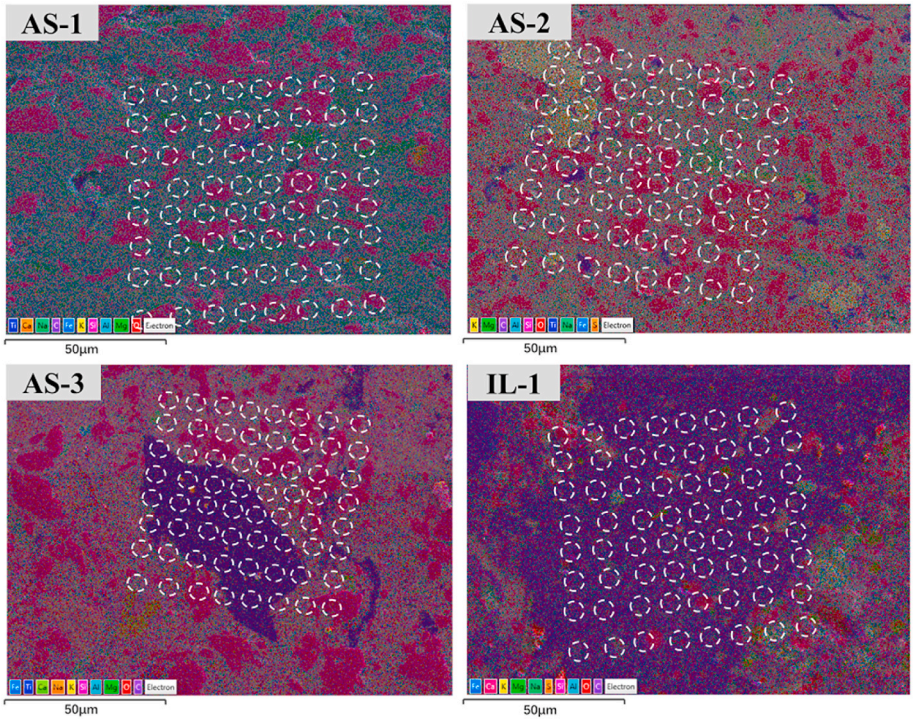


Fig. 8. EDS elemental mapping of the regions on tested shale samples. The elemental distributions of O, Al, Si, Mg, Fe, S, Na, K, and Ti are displayed. The abundance of the elements reflects the mineral composition of the corresponding indentation areas.

matter or a large-grained particle, while plenty of the indents fall on fine-grained mineral regions or grain boundaries. The minerals under the indents should be further partitioned for the detailed property-to-mineral-dependent analyses.

3.3. Identification of the minerals under the specific area

It is vital to analyze how the mineral compositions determine the mechanical behavior of shales. Quantitatively, the mineralogy and its compositional distribution can apparently control the bulk mechanical properties of shale, such as the modulus and hardness. The micro-mechanical property of shale is expected to be correlated to the mineralogy and microstructure of shale and this is the focus of this study. To arrive at the site-specified correlation, it is particularly important to accurately identify the minerals on the residual impression site.

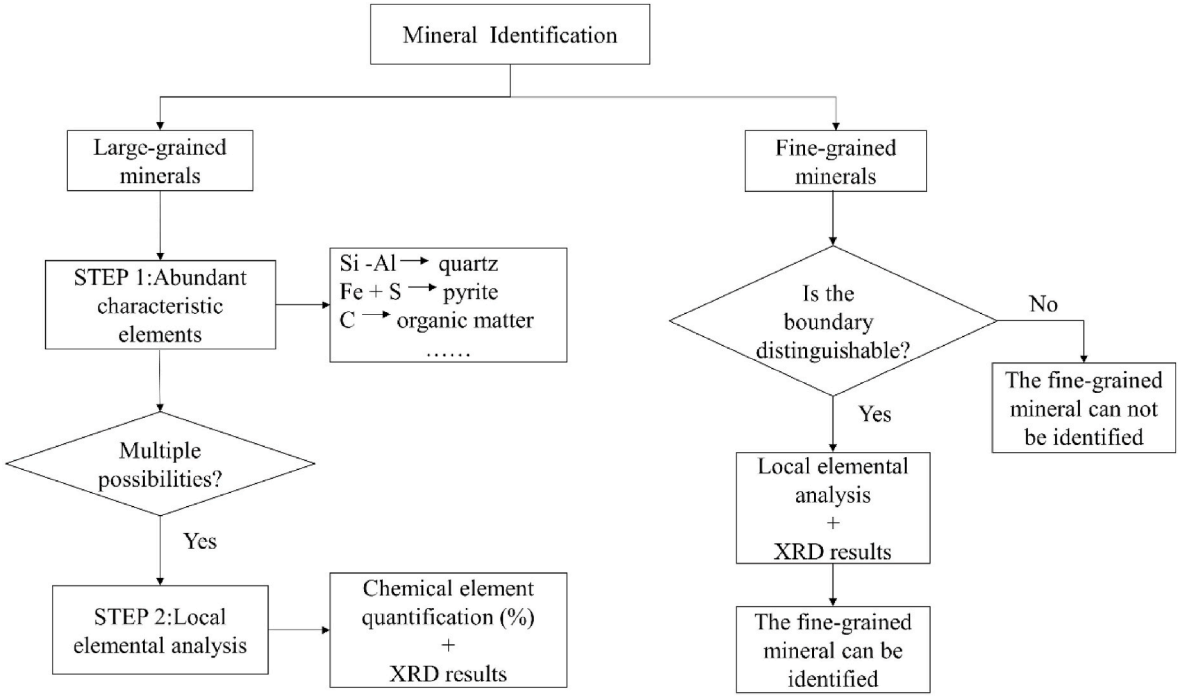


Fig. 9. Schematic diagram of workflow for mineral identification under the indented area on shale sample.

We put forward a practical procedure to incorporate EDS elements analysis with XRD results for mineral identification, as schematically illustrated in Fig. 9. It consists of the identification of large-grained minerals that have large particle sizes or aggregate in a block, or fine-grained minerals that are inserted in the matrix. The identification of the large-grained minerals can be divided into two steps. Firstly, the minerals were identified based on the characteristic chemical elements unique to minerals. Fig. 10 gives an example of the mineral identification in sample AS-2. First, SEM imaging was used to find the residual impression matrix, then, EDS elemental analysis was performed on this target area and the elemental overlapped image was obtained. As shown in Fig. 10, the EDS layered image was divided into different mineral areas, manifesting as different colors. The highlighted areas in magenta are Si-rich large-grained minerals. These areas also show the absence of the Al element and thus exclude other silicates and are identified as quartz. The bright purple areas are carbon-rich organic matter and the regions where the orange S element and the blue Fe element overlap can be recognized as pyrite. This step can identify organic matter and most of the large-size particles. In the second step, the remaining large-grained minerals that cannot be identified in the first step are determined by local element quantification. Take sample AS-2 in Fig. 10 as an example, in the lower right corner of the mapping of the Fe element, two gathering blocks with abundant Fe elements were observed, with the absence of Si, Al, and S in the areas. In fact, there are multiple minerals that are iron-containing. The most effective way is to perform local elemental analysis on the targeted regions. The EDS spectrums on the selected zones, as the 'Spectrum 1' in SEM imaging shown in Fig. 10, indicated that the area is abundant in Fe, C, and O (iridium was from the coating). Combined with the XRD results, it was determined that the blocks were Siderite. After these two steps, the minerals on the shale surface in aggregate form can be accurately identified.

It is challenging to distinguish and recognize the fine-grained layered silicates. The phyllosilicate minerals in the samples are mainly muscovite, clinochlore, kaolinite, and illite. These fine-grained layered silicates are often mixed with each other and have highly variable morphology, depending on the stacking way of their silicate sheets.[70] However, the indenter probe still has chances to contact the pure layered silicate minerals. Therefore, as shown on the right of Fig. 9, another procedure can be executed to determine whether these fine minerals can be identified. For some of the fine-grained minerals with moderate size and obvious boundaries, their element composition can be obtained qualitatively and quantitively by local EDS analysis, which is the same as the second step of large-grained mineral identification mentioned above. To identify the mineral phase, bulk XRD results were used as the reference to compare the relative content of elements. Fig. 11 shows the mineral identification of sample AS-1. Some relatively dense or compacted muscovite and clinochlore can be identified by local EDS analysis owing to their clear boundary. For the ultra-fine clay minerals like illite and kaolinite, it was hard to distinguish them from each other because of the mix of the clay particles and the limited detection area. All the nanoindentation measurements carried on clay were most likely to be the response of their mixture. To summarize, the representative mineralogy of the material under each nanoindentation can be identified in a coarse-grained area and a part of the fine-grained layered silicates within the shale matrix.

For the residual impressions that completely fell on a single mineral particle, as shown in Fig. 12, the identified mineral components were classified into one of the following five mineral categories: 1-quartz; 2-organic matter; 3-muscovite; 4-clinochlore; and 5-iron-type minerals. The iron-type minerals include pyrite, ilmenite, siderite, and some iron oxide, and they usually exist as framboids or incomplete aggregate. The rest of the indents that did not fall entirely or partly on a single-component mineral particle were likely to be on fine-grained clay minerals or cement, which were distributed in the matrix and distinguished unable by statistical nanoindentation tests. Therefore, these indents were grouped into the additional group as 'mineral assortment

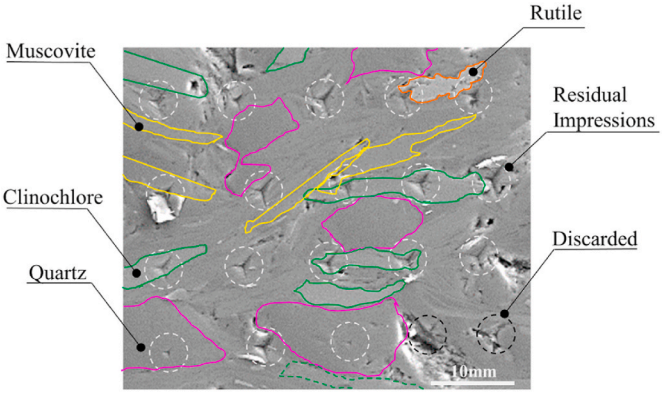


Fig. 11. Example of the minerals identified on the SEM image (sample AS-1). Regions bounded in magenta are quartz. The stripped and band-shaped minerals are layered silicates including muscovite and clinochlore. Indents that fell on rough surfaces are discarded in the analysis.

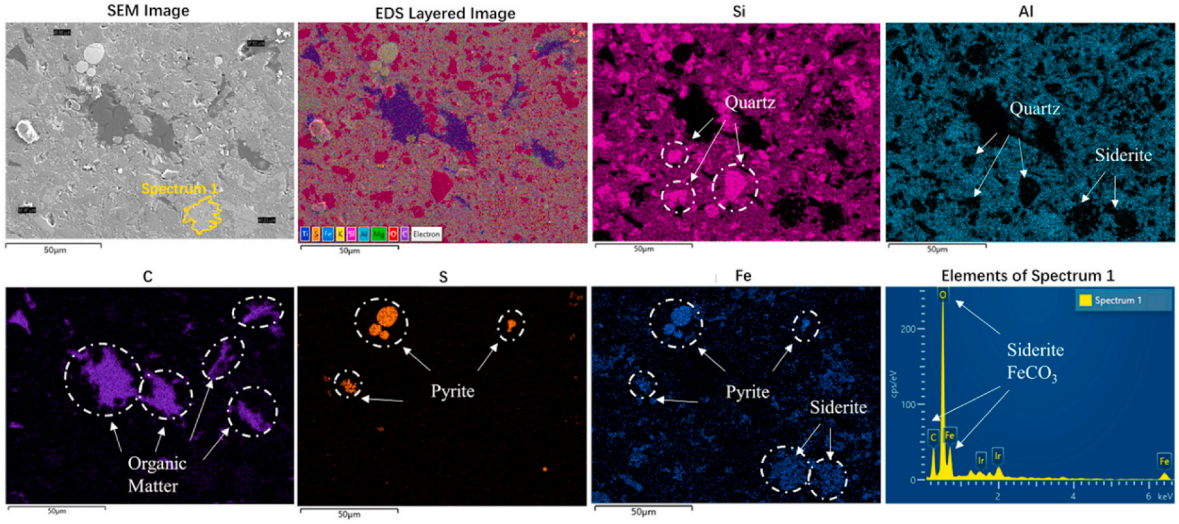


Fig. 10. SEM image, EDS layered image, and the EDS element mapping of Si, Al, C, S, Fe of one of the indentation areas on shale sample AS-2. The last picture is the local element composition of Spectrum 1 by EDS. The minerals under the residual impressions can be identified through SEM-EDS images and XRD results.

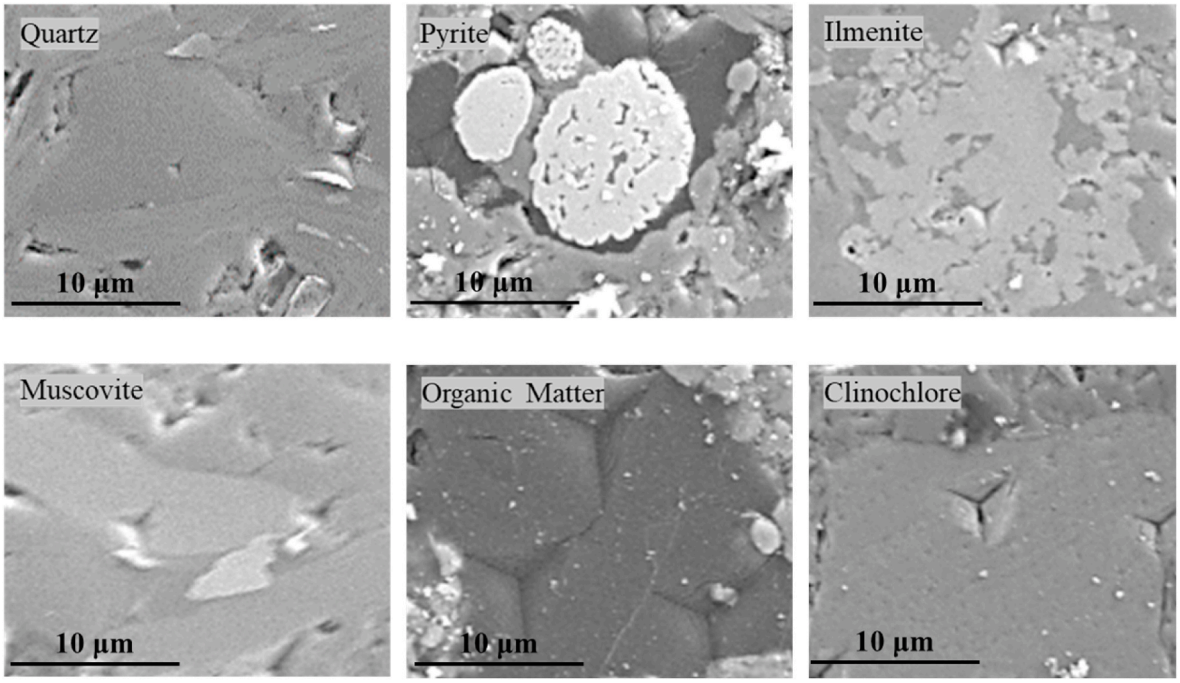


Fig. 12. SEM images of the residual impressions fell on quartz, pyrite, ilmenite, muscovite, organic matter, and clinochlore. The pyramidal residual impressions are small in hard minerals like quartz, pyrite, and ilmenite, while the impressions on relatively soft minerals such as organic matter and clinochlore are large. Since the indentation load is the same, the indentation depth reflects the hardness of the mineral being measured.

in the matrix'. In addition, we also have some indentations in minerals with minor content, such as rutile and apatite, which were not classified into any category because of their trace amount.

3.4. Mechanical properties of minerals in shale

Fig. 13(a) shows the typical load-displacement curves of the five minerals groups. The mineral groups are different in the curve patterns and the maximum depth, indicating the different mechanical responses to the indentation load. As shown in Fig. 13, the load-displacement curves of quartz and iron-type minerals are smooth, and they are steep in the loading process, short in holding creep, and shallow in residual impressions, indicating the high stiffness and hardness of these two types of minerals, which is further verified in the later of this section. For muscovite, clinochlore, and organic matter, the creep in the holding stage, the maximum depth, and the residual depth become larger, especially in organic matters who exhibited a most gently loading process and deepest in depth (over 1000 nm). This manifests that their stiffness and hardness are obviously weaker than the iron-type minerals and quartz. Among the typical load-displacement curves, the creep deformations in the holding stage for the five mineral groups (iron-type, quartz, muscovite, clinochlore, organic matter) are 6,4,8,18,30 nm, and the residual depths are 58,132,181,360,537 nm, respectively, as the line charts in Fig. 13(b) and (c). In addition, abnormal sudden slope changes in the curves which are known as 'pop-in' events,[71] were frequently observed in relatively soft minerals, especially in muscovite curves, which may owe to their unstable layered microstructure.

The elastic moduli and hardness of the mineral groups can be

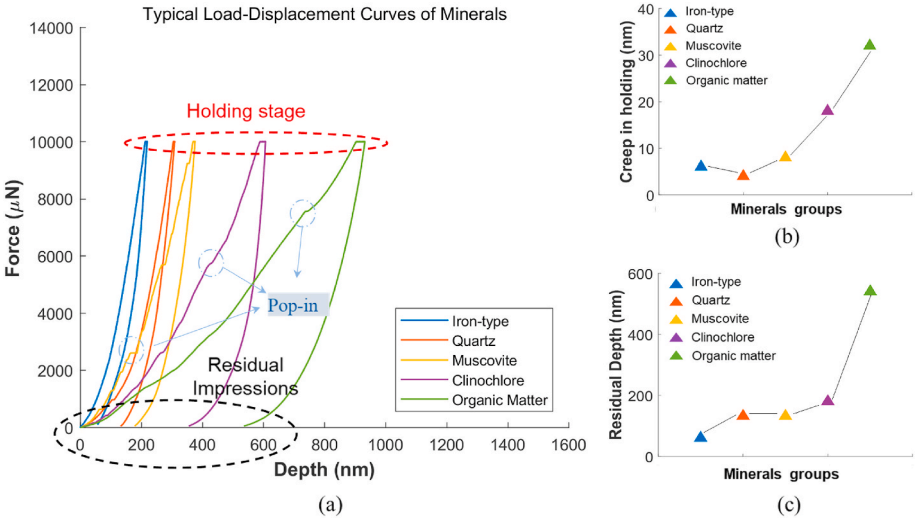


Fig. 13. (a) Typical load-displacement curves of quartz, iron-type minerals, muscovite, clinochlore, and organic matter; (b) The creep displacement in the holding stage of the five mineral groups; (c) The residual depth of the impressions on the mineral surfaces after the indenter tip withdraw.

extracted from the load-displacement curves. The elastic modulus E measures the resistance of the mineral to the elastic deformation in the indentation experiment, while the hardness H manifests the resistance to the plastic deformation. The elastic moduli and hardness of the identified mineral groups are shown in Fig. 14. The values of each mineral phase are present as a boxplot. On each box, the central, bottom, and top lines indicate the median, the 25th, and 75th percentiles, respectively. The black circles indicate the average values, and the red points are the outlier.

As shown in Fig. 14(a), the iron-type minerals showed the highest elastic modulus, followed by quartz, muscovite, clinochlore, and organic matter. As to the hardness, as shown in Fig. 14(b), quartz was harder than iron-type minerals, and then followed by muscovite, clinochlore, and organic matter. The elastic moduli of iron-type minerals were observed to be up to 167.0 GPa, owing to their stable and compact crystal structure. The elastic moduli and hardness of quartz ranged from 43.4 to 116.8 GPa and 4.4–18.4 GPa, which were consistent with the previous studies.[32,47,72] The values of iron-type minerals are more scattered than those of quartz due to the multiple kinds of minerals in the iron-type group. Following quartz, muscovite showed moderate modulus and low hardness, with an average value of 55.4 GPa in modulus and 2.7 GPa in hardness. The clinochlore showed an average modulus of 37.0 GPa and hardness of 1.5 GPa. The measured modulus of organic matter ranges from 7.9 to 26.1 GPa and is in good agreement with the reported value.[51,73,74] Previous study shows that the shale possesses multiscale anisotropy.[70] The orientation of the particles will influence the obtained mechanical properties. For instance, clay minerals may be expected to show a higher stiffness when indented parallel to the bedding plane compared to perpendicular to the bedding plane.[75] The different orientations of the minerals may be a reason for the dispersive mechanical data. How the anisotropy at the nanometer scale affects the mechanical properties of shale is a topic of interest in the future.

To relate the mineralogy to the mechanical responses, we calculated and compared the related energy induced in the indentation test. The area enclosed by the load-displacement curve and the X-axis represents the energy dissipation during the nanoindentation test, which reveals the elasto-plastic properties of the indented mineral. As illustrated in Fig. 15, the total work W_t of the curves was calculated by the area of the irregular polygon enclosed by the loading curve, holding stage, and X-axis. Then the plastic work W_p was obtained by subtracting the elastic energy (the area enclosed by the unloading curve and X-axis) from the total work. It was noted that the highly concentrated indentation force may cause the brittle failure of the minerals and lead to potential microcracks on the surface. Thus plastic work is composed of two parts:

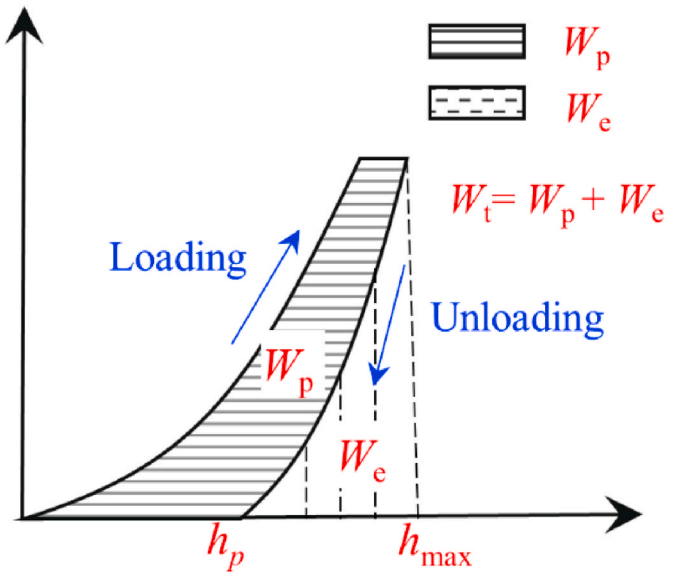


Fig. 15. Schematic of the related energy induced in the nanoindentation test. W_p denotes plastic work, W_e denotes elastic work, and W_t is the total work.

pure plastic work that dissipates due to pure plastic deformation, and energy lost due to induced cracks.[76] Since it is impossible to know the number and the length of the induced microcracks, the energy dissipated in the crack initiation and propagation was not considered. Thus, the plastic energy and the total energy ratio reported here have implicitly included of the potential cracking energy.

Results are shown in Fig. 16. It was observed that the minerals demonstrate different elasticity- or plasticity-preferred characteristics, related to the crystal structure of the mineral. Silicates minerals with the framework crystal structure (tectosilicates) like quartz exhibit the lowest plastic work with an average value of 384.1 pJ, while phyllosilicates like muscovite and clinochlore exhibited quite large plastic work over 800–2000 pJ. Iron-type minerals showed a wide range in the plastic work which may be owing to the different elasto-plastic behaviors of the multiple minerals in the group. As expected, the plastic work of organic matter was up to over 2500 pJ, attributing to the significant compliance and softer properties than other inorganic mineral components. In order to quantify the contribution of plastic behavior for each mineral group during the indentation process, the ratios of plastic work to the total work of the mineral groups were calculated and shown in Fig. 16(b). The results showed that quartz was dominant in elastic behavior during the

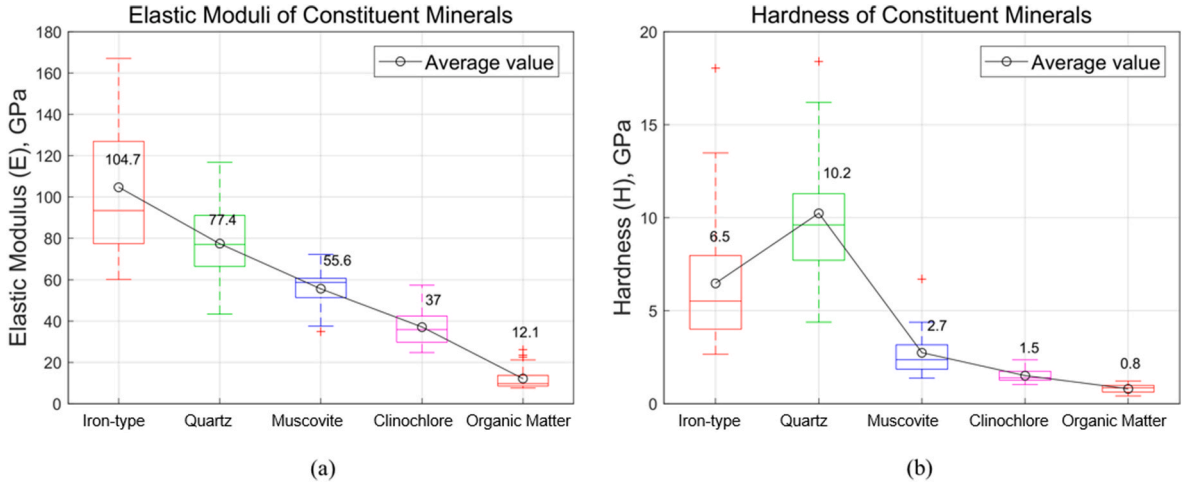


Fig. 14. Statistical analysis in box plot of the (a) elastic moduli and (b) hardness of the identified mineral phases.

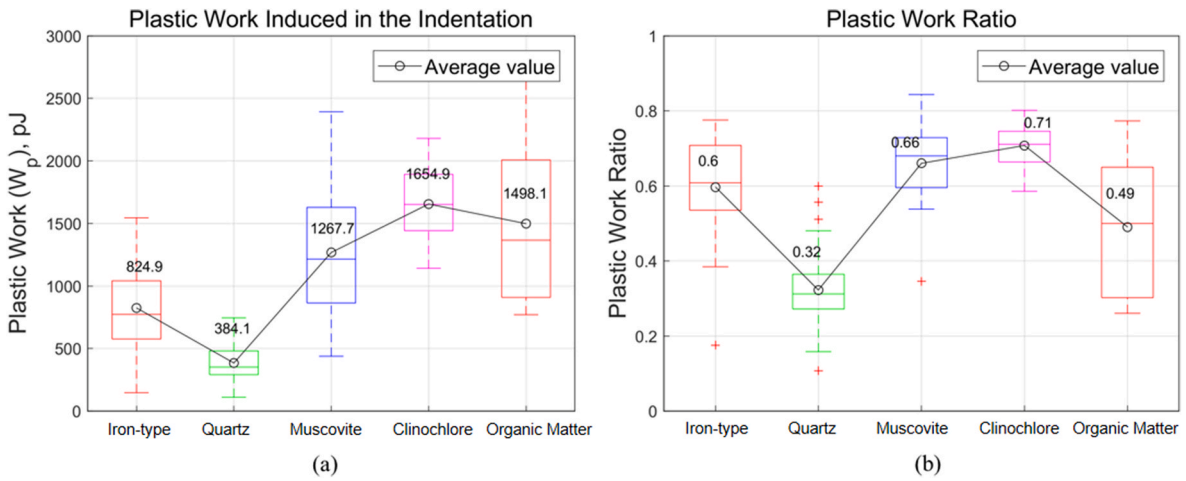


Fig. 16. Statistical analysis in box plot of the plastic work induced in the indentation test (a) and the plastic work ratio (b) of the identified mineral phases.

indentation and its average plastic ratio was lowest at 0.32. Plastic behavior played a vital role in the mechanical responses of muscovite and clinochlore due to their orientation-preferred stacking arrangements. As for iron-type minerals, the plastic behavior was also obvious. It was interesting that the organic matters showed both elastic-dominant and plastic-dominant behaviors to the indentation load, with the plastic ratio ranging from 0.3 to 0.8. This may be related to the chemical compositional differences and various thermal maturity of kerogen in the shale samples. The thermal maturity of the kerogen in AS-2 and AS-3 is high while IL-1 shale has a fairly low thermal maturity, thus may be different in elastic properties. Previous studies found that mature and overmature kerogen particles have higher stiffness.[34,45,51] Different kinds of macerals in kerogen transformed during thermal maturation with physical and chemical alteration.[34] During the maturation, the aliphatic C–H bonds decrease, and aromatic compounds increase, increasing the density of kerogen, which leads to stiffer mechanical responses.[45] With the increase of thermal maturity, the organic matter becomes more structurally heterogenous, which changes the porosity in kerogen and also leads to different mechanical behaviors. In addition, the tested value may be influenced by the adjacent minerals and or the mineral under the tested area, known as the surround effect and substrate effect.[47,70,72,77] It was found that the penetration depth had an impact on the measured mechanical properties.[12,44,78,79] Indentations at shallow depth are more likely to fall on a single mineral phase and provide measurements of corresponding constituents, while at deep depth, the probe may penetrate multi constituents under the surface and thus returned the mechanical properties of the composite which contains the contribution of surrounding minerals and supporting constituents.[12,20,80]

Table 2 summarizes the statistical average properties of the identified minerals. The result of this work demonstrates that the unique mechanical properties and distinct elasto-plastic responses of the rock-forming mineral grains, further proved the strong heterogeneity of micromechanics of shale rock. The probing mechanical properties on nano-scale related to the evaluation of the ductile behavior of shale at mineral level. This can contribute to the prediction of the initiation and propagation of microcracks as well as the potential failure of shale, which are important for the permeability improvement and achievement on economical gas flow. Besides, it also involves the proppant embedment during the treatment. Exposure to hydraulic fracturing fluids will change the mechanical properties of shale formation, such as Young's modulus. This leads to weakening of the rock frame, which has the potential to increase proppant embedment into the fracture face, resulting in reduced conductivity.[81]

Table 2
Statistical average properties of the identified mineral phases.

	E (GPa)	H (GPa)	W_p (pJ)	W_p/W_t
Iron-type minerals	104.7	8.5	824.9	0.60
Quartz	77.4	10.2	384.1	0.32
Muscovite	55.6	2.7	1267.7	0.66
Clinochlore	37.0	1.5	1654.9	0.71
Organic matter	12.1	0.8	1498.1	0.49

3.5. Mechanical properties of the mineral assortment in matrix

The mechanical properties of the crystals or amorphous phase represent the smallest scale thereof being studied. Apart from the five identified mineral groups presented above, more indents did not fall on a single-component mineral particle. These indents fell on the assortment of fine-grained mass material which includes clay minerals, clastic particles, and the interparticle cementing substance. The mechanical behavior of mineral assortments in the matrix also plays a critical role in the macro behavior of shale since it serves as the continuous skeleton of the shale matrix. The crystals and amorphous inclusion in shale are integrated by various cementing materials such as quartz, carbonate, iron oxides, and clay minerals.[82] These cementing materials precipitate locally on grain or pore spaces during the diagenesis, binding the sediment grains together, causing a decrease in porosity, and altering the rock stiffness.[83] We extracted the probing data apart from the ones on skeletal grains and grouped them into additional group as a mineral assortment in the matrix. As shown in Fig. 17, three kinds of mineral assortment in the matrix were observed in the samples. One is phyllosilicate-rich assortment, which was mainly the mixture of fine-grain silicate minerals such as muscovite, illite, kaolinite, as well as a few other particles. Strips or bands of minerals with preferred orientations were observed in the matrix, as shown in SEM Image 1 in Fig. 17 (a). This assortment was rich in elements like Mg, Al, Na, and other common elements in phyllosilicate minerals. Second is the iron-mineral-rich assortment, including small particles of iron-type minerals distributed in the assortment instead of showing as clusters, as shown in SEM Image 2 in Fig. 17. This assortment in the matrix was rich in Fe element. And the third is the organic-rich assortment, which was rich in kerogen and a small amount of isolated granular minerals, as shown in SEM Image 3 in Fig. 17. The content of the C element was abundant in the EDS image in this kind of assortment.

Fig. 18 shows the statistical analysis in box plots of the mechanical properties for three types of assortments. The estimated elastic moduli and hardness of iron-mineral rich assortments were significantly higher than the phyllosilicate assortment and organic-rich assortment. The iron-type particles serve as the rigid skeleton in the shale matrix, even if

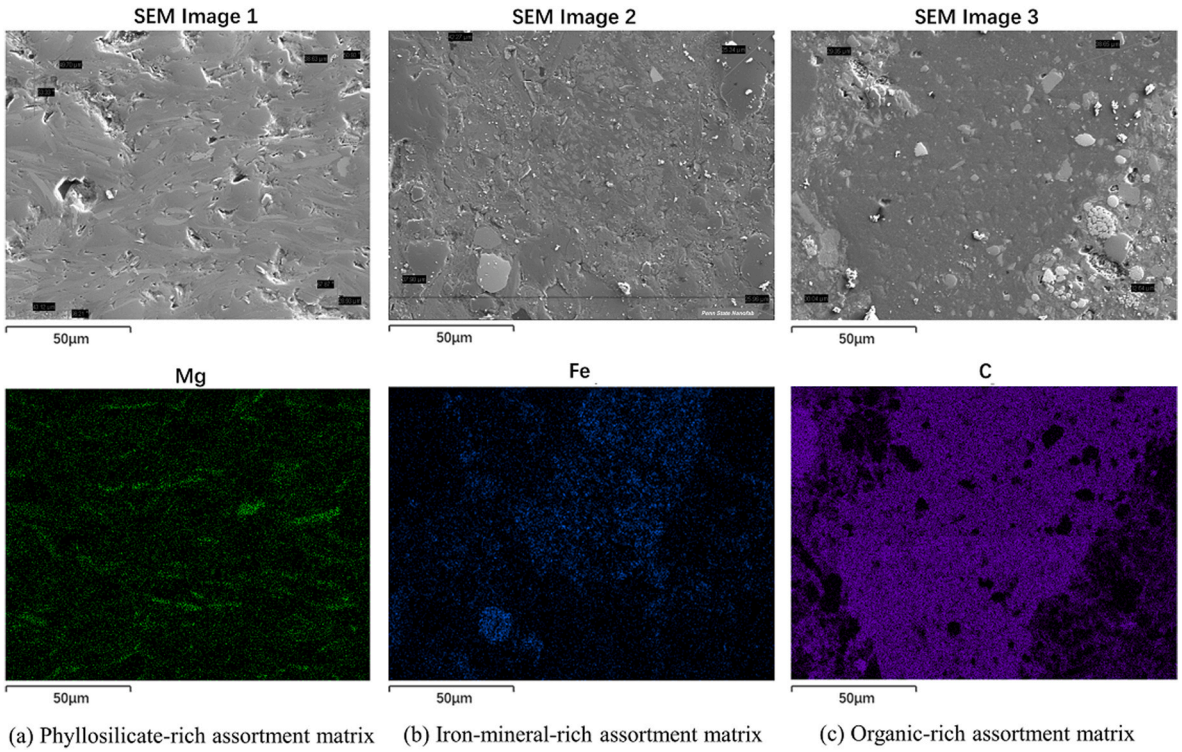


Fig. 17. SEM images and EDS images of the shale sample AS-1 (Image 1), AS-3 (Image 2), and IL-1 (Image 3). They represent three different kinds of mineral assortment in the matrix: (a) Phyllosilicate-rich assortment, which was mainly the mixture of fine-grain silicate minerals such as muscovite, illite, kaolinite, as well as a few other particles; (b) Iron-mineral rich assortment, including small particles of iron-type minerals distributed in it; (c) Organic rich assortment, which was occupied mainly by kerogen and a small amount of isolated granular minerals inserted in it.

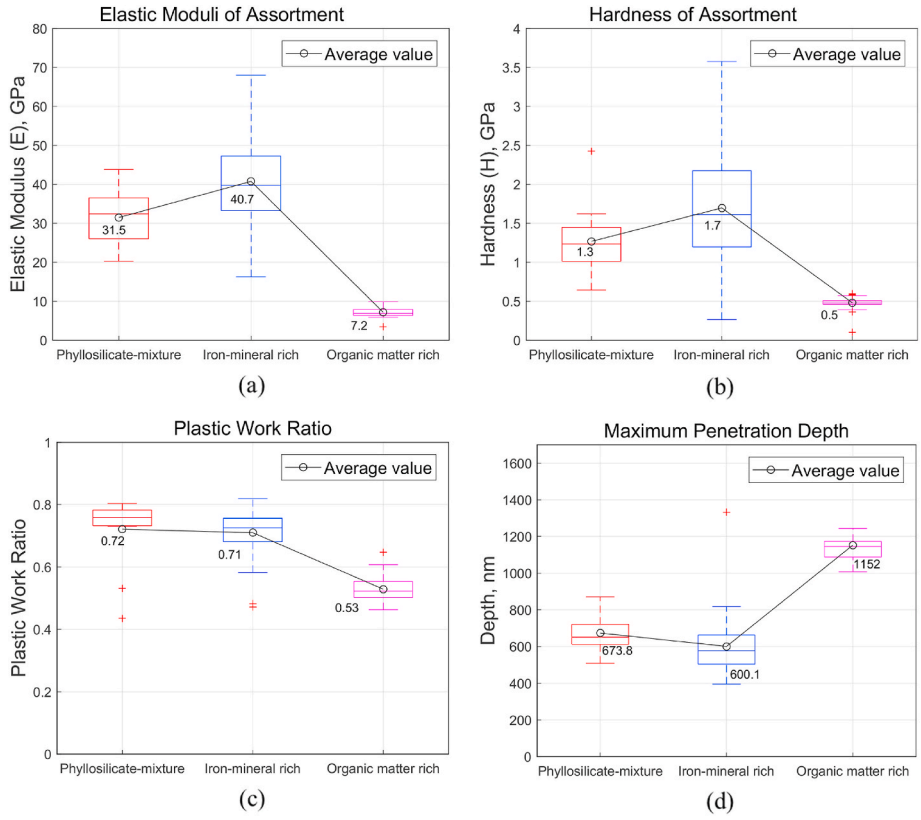


Fig. 18. Statistical analysis in box plot of the (a) elastic moduli, (b) hardness, (c) the plastic work induced in the indentation test, and (d) the plastic work ratios of the identified assortments that are rich in phyllosilicates, iron-minerals and organic matter, respectively.

they were in small size. The large range of the mechanical properties of these particles manifested the high heterogeneity of the assortment in the matrix. Phyllosilicate-rich assortment showed moderate elastic moduli and hardness, and the most significant plastic behavior. This indicated that the micrometer-sized sheet-like clay minerals and other layered silicates (muscovite) played a dominant role in the overall plastic behavior of shale. The organic rich-assortment showed the lowest moduli and hardness. The mechanical properties of organic-rich assortment were more narrowly distributed than the other two types of assortments as shown in the pink box plot in Fig. 18. This demonstrates that it was considerably homogenous. Interestingly, the plastic work ratios of organic-rich assortment were about 0.45–0.6, suggesting that the elastic response played an important role in the deformation of the organic-rich assortment. This may be attributed to the contribution of surrounding minerals and based-supporting constituents. As it can be seen in Fig. 18(d), the penetration depths of the indents fell on organic-rich assortment are over 1 μm, much deeper than the phyllosilicate-rich assortment and iron-mineral rich assortment. Thus, it was very likely that the indentation measured the composite response which contains the contribution of their surrounding and based minerals. The statistical properties of the mineral assortments in matrix are summarized in Table 3.

In addition, we found that the measured mechanical properties of the inorganic mineral particles in the organic-rich assortment were far below the aforementioned values. The measured moduli of pyrite and quartz in Sample IL-1 were only 9.5–38.6 GPa and their hardness were 0.6–1.35 GPa. Sample IL-1 was collected from the coal roof rock, and XRD-EDS results confirmed that it has a very high content of organic matter. As shown in Fig. 17, the hard minerals are isolated and 'floated' in the organic assortment. The discrepancy between the measurements and the expected values is very likely result from limitations in the Oliver–Pharr method for determining contact area. The determination of the mechanical properties of the individual mineral grain was influenced by the substrate underneath the mineral particles. The quartz or pyrite grains were isolated in the compliant organic matters, which was equivalent to measuring the mechanical properties of a stiff particle on a soft substrate. The substrate yields during the penetration and adds extra depth to the measurement. This extra depth is not considered in the Oliver-Pharr method, which causes an overestimation of the contact area and hence underestimates the elastic modulus and hardness.[84] Similarly, the true contact depth can be underestimated in the case of measuring a soft particle on a hard substrate and thus leads to the overestimation of the elastic modulus and hardness. The mis-measurements can be overcome by taking the influence of substrate compliance into account when calculating the projected area.[47,77,85,86] Explicit consideration of the substrate effect is particularly important in the measurement of clay mineral particles because their thickness often presents less than 1 μm. This increases the difficulty of the mechanical measurements of the clay minerals. An effective model should be developed to study the mechanical properties of clay minerals on the sub-micron scale in the future.

3.6. Indentation damage in shale: the cracking patterns

When conducting indentation tests on shale, the minerals on the polished shale surface that encounter the indenter tip can be mechanically stiff or soft. The inherently porous-structure and multi-composition have a significant influence on the mechanical response of a specific local volume below the indenter. Although cracking events should be carefully avoided in the nanoindentation test, the occurrences of cracks are almost inevitable when conducting indentation test on siliceous shale. Three cracking patterns were observed in the indentation tests on shale: the lateral cracks, radial cracks, and shear cracks, as illustrated in Fig. 19. The lateral cracks were formed near the center of the indenter tip and extended to lateral cracks, while the radial cracks initiated from the location where stress concentrates such as the indenter corners. The shear cracks usually occurred around the triangular indents and formed discontinuous shear bands around indents. The SEM images of residual impressions showed that the cracks frequently occurred on the grain boundaries as well as soft mineral phases, as shown in Fig. 20(a)-(c). The indents fell on the boundary of minerals with high strength mismatch were prone to induce the radial cracks, such as the boundary of quartz-matrix or quartz-clay, which may be attributed to stress concentration and the uncoordinated deformation between the hard mineral and soft mineral. It was observed that radial cracks were mainly found on hard minerals like quartz, as shown in Fig. 20 (a)–(c), while the shear cracks often occurred on relatively soft minerals like layered silicates and organic matter, as shown in Fig. 20(b), (c). Combining the SEM images with the corresponding load-displacement curves, it was found that the formation of the radial cracks usually corresponds to a slight pop-in event, as shown in Fig. 20(a) and (b). The formation of the shear cracks, which are great in numbers, are accompanied by a series of abnormal slope changes as shown in Fig. 20(c). On the contrary, Fig. 20 (d) exhibits an undamaged surface on quartz where no crack was induced in the nanoindentation. It was observed that the corresponding load-displacement curves are quite smooth and perfect. The pop-ins on the load-displacement curves indicate that the initiation of cracks brings a sudden increase in depth which may be an extra penetration of the indenter. This behavior relates to the microstructure in the contact area.[57]

The crack formed in the indentation tests evolves with the loading process. Previous studies on indentation tests on other brittle solids, such as glass and ceramics, also found that median (radial) cracks and lateral cracks were formed in the deformation zone.[87,88] Cracks initiate from the deformed plastic zone (the hemispherical area), as shown in Fig. 21. The median cracks are formed in the loading process while lateral cracks are formed in the unloading process. It is believed that the median crack initiates below the contact point when the stress concentration surpasses the critical point and then undergoes a stable crack growth. The median crack begins to close as the unloading process progresses. With the withdrawal of the indenter, the deformed material in the contact zone re-gains its shape and the relaxation superimposes intense residual tensile stress upon the applied field. That is when the lateral cracks appear. For a highly heterogeneous, natural material such as shale, the influences of the microstructure on the crack initiation are ought to be taken into consideration. In the initial loading stage, the natural microcracks and some pores below the contact area gradually closed under the high pressure of the probe. In this stage, the cracks are less likely to form because of the non-obvious stress concentration at the corners. Subsequently, irreversible deformation occurs when the loading goes in elasto-plastic process. The unbalanced energy between the hemispherical plastic zone forms beneath the indenter tip and the outside elastic zone could arouse the initiation of cracks. If there are natural flaws in the vicinity of the contact area, lateral cracks might form in the loading stage and even connect with radial cracks, causing chipping in brittle minerals, as shown in Fig. 22(a). In addition, the formation of shear cracks may be triggered because of multiple reasons. First, it is related to the arrangement of sheets structure of layered silicate. Fig. 22 (b) shows the SEM image of the stacking sheets of the layered silicates. The stacking disorder of the silicate sheets may result in dislocation and slip under the pressure of the indenters, resulting in shear cracks. Second, it is related to the shear bands resulting from the severe deformation of the indented surface in case of the indenter contact with an amorphous phase.[89,90] Fig. 22(c) shows a residual

Table 3
Statistical properties of the mineral assortments in matrix.

	E (GPa)	H (GPa)	W_p/W_t	$h_{max}(nm)$
Phyllosilicate-rich	31.5	1.3	0.72	673.8
Iron mineral rich	40.7	1.7	0.71	600.1
Organic rich	7.2	0.5	0.53	1152.0

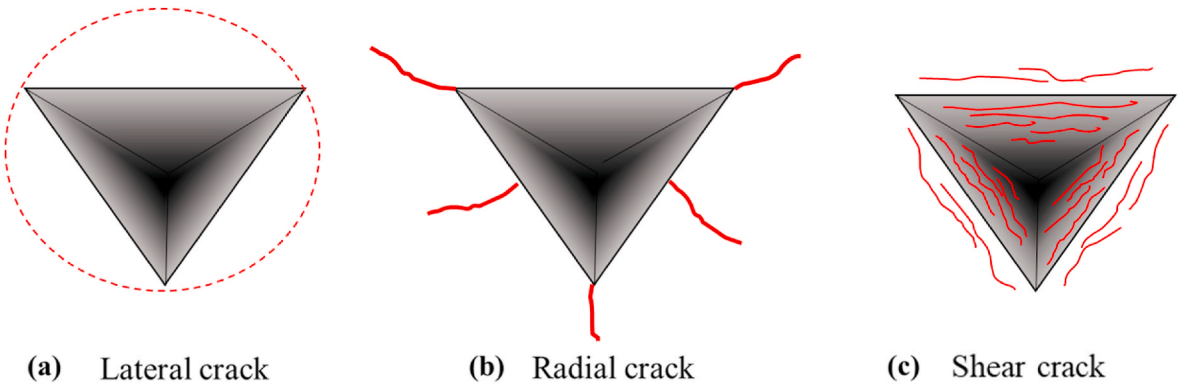


Fig. 19. Schematic of three cracking patterns that were observed in the indentation tests on shale: (a) lateral crack, (b) radial crack, and (c) shear crack.

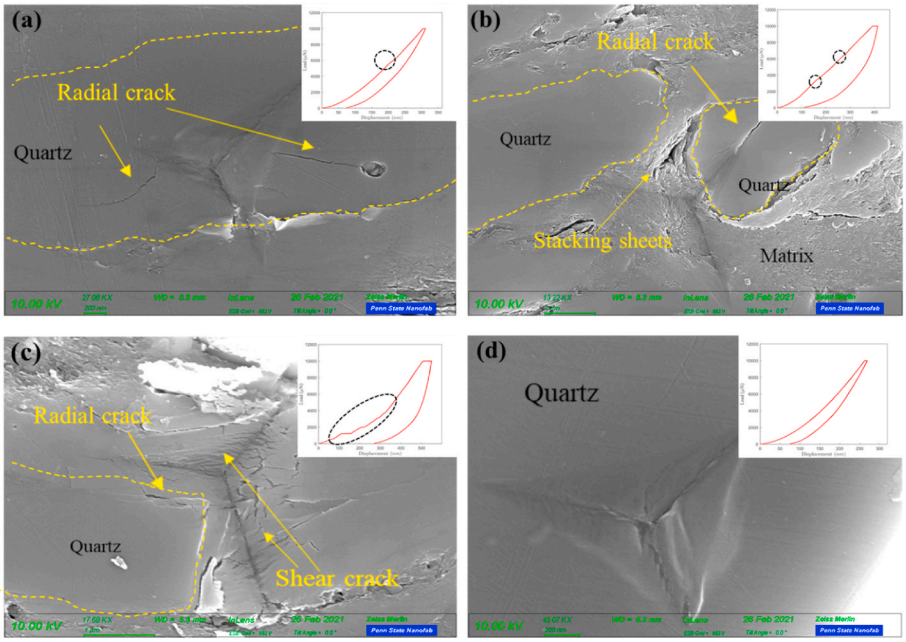


Fig. 20. SEM images of residual impressions induced in the nanoindentation tests and the corresponding load-displacement curves. Radial cracks were observed on the quartz, and shear cracks were formed on the compliance minerals. Abnormal changes were observed in the load-displacement curves of the indents that induced cracks on the surface, as shown in (a), (b), and (c). The abnormal change was not observed in the displacement-load curve of the undamaged intended surface, as shown in (d).

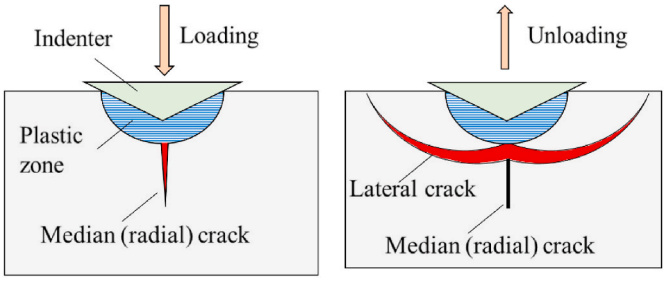


Fig. 21. Schematic of the crack deformation under the indentation. The median cracks are formed in the loading process (left) when the stress concentration surpasses the critical point. The lateral cracks appear in the unloading process (right) owing to the intense residual tensile stress due to the relaxation of the deformed material in the contact zone.

impression fell on organic matter in sample AS-3. The intense shear bands along the sides of the pyramidal residual impression bounded within the triangular indent are obvious.

4. Summary and conclusion

In this study, nanoindentation tests were conducted on four shale samples with different mineralogical properties. The representative mineral groups in shale were identified with the assistance of SEM-EDS-XRD analysis and their probing mechanical properties were obtained by nanoindentation. We analyzed the mechanical properties and deformative behavior of the quartz, iron-type minerals, muscovite, clinochlore, and organic matter. The three kinds of assortment were mechanically evaluated. We discussed the mechanism of the induced cracks in tests. Based on the results, the following conclusions can be drawn:

(1) The identified mineral groups are apparently different in terms of their estimated mechanical properties on the tested scale. The elastic moduli scope (average values) ranking in descending order are: iron-type minerals 60.1–167.0 GPa (104.7 GPa), quartz 43.4–116.8 GPa (77.4 GPa), muscovite 34.8–72.2 GPa (55.6 GPa), clinochlore 24.7 GPa–57.3 GPa (37.0 GPa), organic matter 7.9–26.1 GPa (12.1 GPa). The hardness (average values) ranking in descending order are: quartz 4.4–18.4 GPa (10.2 GPa), iron-type minerals 2.7–18.0 GPa (6.5 GPa), muscovite 1.4–6.7 GPa (2.7 GPa), clinochlore 1.0–2.4 GPa (1.5 GPa), organic matter

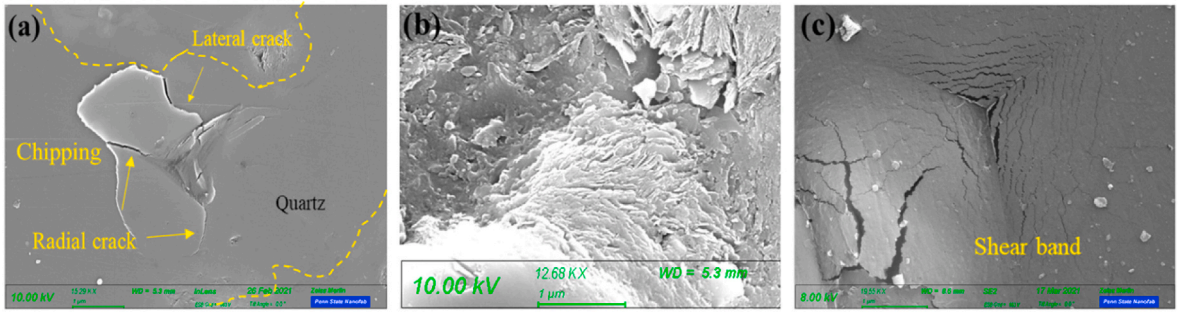


Fig. 22. (a) Chipping occurs in brittle minerals after cracks were induced. (b) SEM image of the stacking sheets of the layered silicates. (c) Pyramidal residual impression of the indent fell on organic matter in sample AS-3. Shear bands resulting from the severe deformation of the indented surface are evident.

0.4–1.2 GPa (0.8 GPa). Note that the above mechanical properties are relevant only in the direction perpendicular to the bedding planes.

(2) The identified minerals demonstrated different elastic- or plastic-preferred characteristics to the indentation load. Quartz exhibits the lowest plastic work, while phyllosilicates like the muscovite and clinochlore exhibit quite large plastic work. Organic matter showed the largest plastic work attributing to the significant compliance and softer properties than any other inorganic mineral components.

(3) The mechanical properties of the mineral assortment in matrices are significantly influenced by their constituent minerals and their distributions. The measured elastic modulus and hardness of iron-mineral rich constituent matrix (40.7GPa/1.7 GPa) were significantly higher than the phyllosilicate constituent matrix (31.5 GPa/1.3 GPa) and organic matter rich constituent matrix (7.2GPa/0.5 GPa). It was found that the measured mechanical properties of the inorganic mineral particles in the organic-rich matrix were far below the values that were measured in the phyllosilicate rich and iron-mineral rich assortment matrices. This is because the substrate compliance caused extra depth that is not considered and thus leads to the overestimation of elastic modulus and hardness.

(4) It was observed that radial cracks are formed when the indents fell on the boundary of brittle minerals like quartz and relatively weak minerals. The high strength mismatch between the hard mineral and soft mineral leads to uncoordinated deformation under indentation load, finally prompting the initiation of the cracks. Shear cracks were observed on layered silicates and organic matters, which are related to the stacking structure and the shear bands resulting from severe deformation, respectively.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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